

Dispositivos Gráficos

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Dispositivos Gráficos

- Dispositivos de Entrada
- Dispositivos de Saída
- Processamento

Entrada

Quais dispositivos nos permitem fazer entrada de dados posicionais (2D e 3D) ?

Teclado

...sim, o velho e bom teclado
QWERTY/AZERTY, DVORAK,...



Mouses Convencionais

(X e Y são próprios do dispositivo)



2 botões base + 4 botões
extras + X, Y + scroll

X, Y + 2 botões + scroll



Mouse Siameses

(X, Y + 2 botões)



Mesa Digitalizadora (2DOF simultâneos + N botões)



Joystick / Manche

$X + Y + Rz + \text{botões}$



Outros Controles



Mouse de Giro

(Gyraton Ultra GT Cordless Optical Mouse)

Sem fio

2DOF - Tem giroscópio de dois eixos (Rx e Ry no ar/espaco e não numa superfície) + botões



Trackball



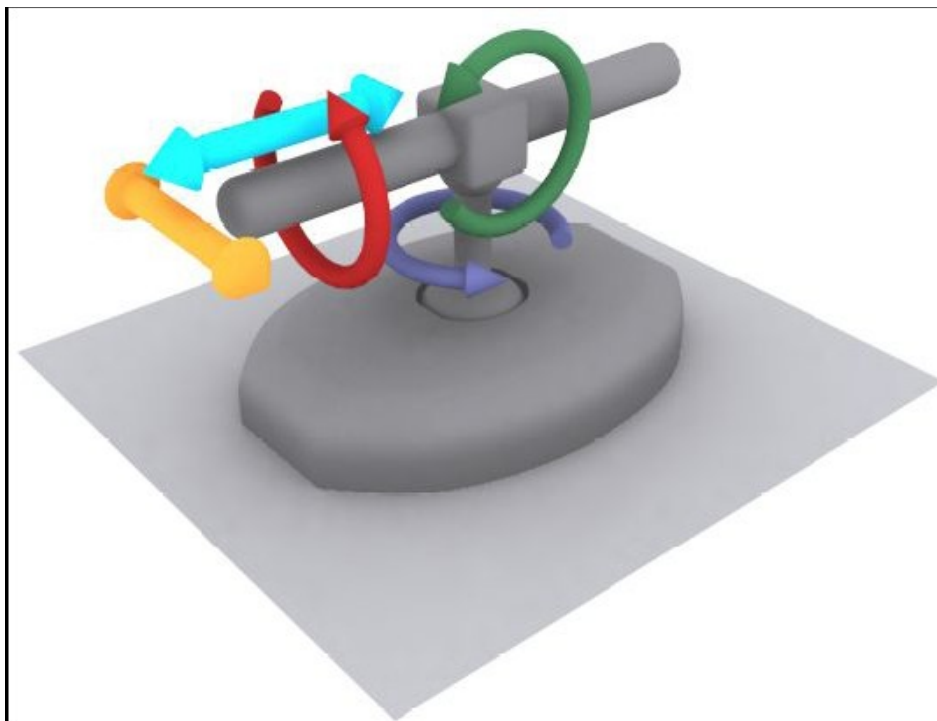
Magellan 3D controller (mouse 3D)



6DOF simultâneos

+ N botões

HandleBar (BARNI, MELLER e NEDEL, 2006)



5 DOF simultâneos

2 posicionais

3 orientacionais

Sistemas de Feedback Tátil



Luvas, DataGloves

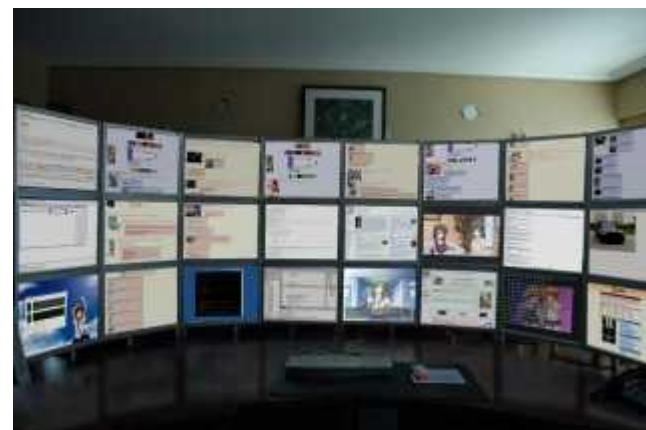


Saída

Projektor



Multi Monitor



Monitores em diferentes formatos



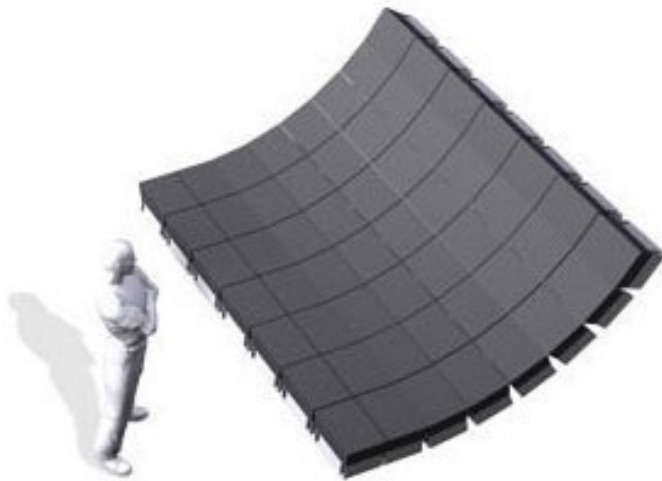
Rectangular Versions



Modular Design: Screen, Base Frame, Projector



Hexagonal Versions

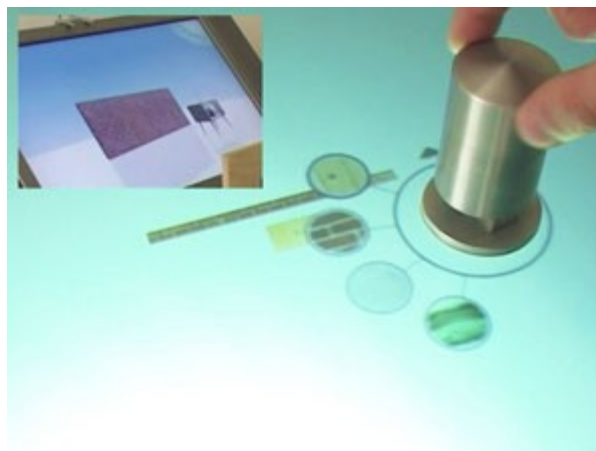


Convex, Concave or Straight Arrangements

VISIO STATION



Mesas Interativas



Immersa Desk



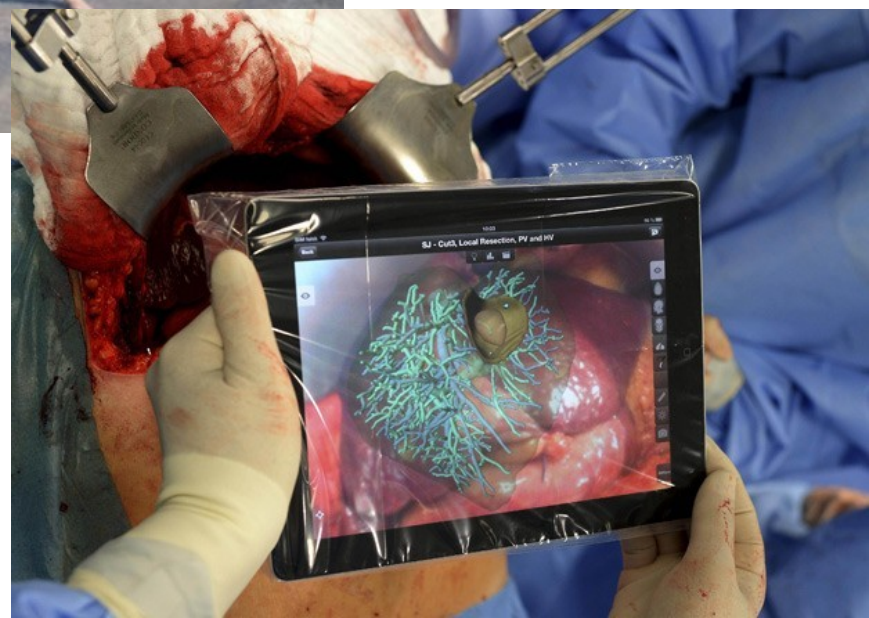
WorkWall



Personal Monitor



Tablet/Smartphone para RA



HMD

Head Mounted Displays

Head Mounted Displays (HMDs)



Head Mounted Displays (HMDs)

Modelo que permite
uso em **Realidade
Aumentada** pois
simula funcionamento
“**Semi Imersivo**”





Virtual Binoculars



Visette45 SXGA



hi-ResSVGA+



Hi-Res800 (3D)



Visette® Pro



Window VR



VR 1280



Ruggedized i-Visor



Ruggedized i-



Z800 eMagin



SimEye SX100



SimEye SX45



SimEye SX50T



MicroView 35



MicroView 35T



nVisor ST50



nVisor ST60



nVisor SX60



nVisor SX111



Virtual Binocular
SX



Virtual Binocular
SV



VRvision HMD



ARvision-S HMD



ARvision 3D HMD



LCD29-HMD



M3 Monocular



M3-Cam



AlphaBino



ARtelescope



3Bino



piSight™



xSight™



zSight™

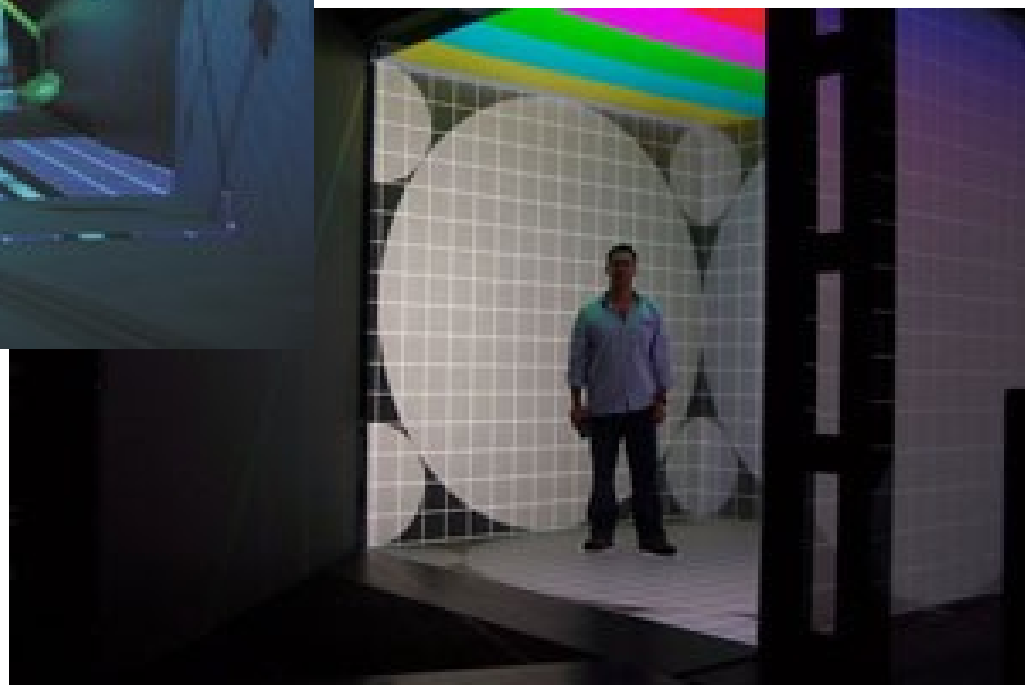
Oculus Rift



Meta Quest 2 e 3 (LARVA)



Cavernas



Processamento



512 KB (640x480 256 cores)



4 MB (1024x768@32Bits)



32 MB (4 shaders, 120 MHz)



24 GB (16384 shading units, 82.58 TFLOPS)